

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIOE604B - CRYPTOGRAPHY & NETWORK SECURITY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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Which of the following is not a principle of security?

- A. Confidentiality B. Availability C. Efficiency D. Integrity

Which attack involves passive listening to communication without altering the message?

- A. Masquerade B. Replay C. Eavesdropping D. Modification

In symmetric key encryption, the same key is used for:

- A. Hashing only B. Encryption and decryption
C. Signature verification only D. Key exchange only

Which of the following is NOT an asymmetric key algorithm?

- A. RSA B. Elgamal C. Diffie-Hellman D. RC4

Kerberos uses which technique for secure authentication?

- A. One-time pad B. Hash chains
C. Ticket-granting system D. Certificate chaining

In IPsec, which component provides encryption of the payload?

- A. Authentication Header B. ESP
C. Security Association D. Tunnel Mode

Which attack does Cross-Site Scripting (XSS) exploit?

- A. Cryptographic flaw B. Input validation flaw
C. Protocol flaw D. Key exchange flaw

Steganography is the technique of:

- A. Encrypting the message using a cipher
B. Hiding the existence of the message
C. Hashing data for authentication
D. Converting plaintext to ciphertext

In Elgamal encryption, the decryption requires:

- A. Private key of sender B. Public key of receiver
C. Private key of receiver D. Shared secret key

- 10 In the TLS handshake, the session key is typically established using:
 A. Symmetric key exchange B. One-way hash
 C. Asymmetric key exchange D. Digital watermarking
- 11 In Single Sign-On (SSO), a user logs in:
 A. Once to access one application only B. Multiple times for multiple apps
 C. Once to access multiple systems D. Each time for each page
- 12 CMAC is a type of:
 A. Digital signature B. Message authentication code
 C. Encryption algorithm D. Key distribution protocol
- Q2. Solve the following.
- A) Define the following terms:
 a) Security Mechanism b) Encryption c) Decryption.
 Differentiate between symmetric and asymmetric key cryptography with appropriate diagrams. 6
- B) Define the following terms:
 (a) Plaintext (b) Ciphertext (c) Steganography 6
 Given a transposition cipher, use the key = 4312 to encrypt the plaintext: 'SECURITY'. Show step-by-step encryption.
- Q3. Solve the following.
- A) Given a system using Diffie-Hellman Key Exchange, Public values: $q = 23$, $\alpha = 5$
 User A selects private key $X_a = 6$
 User B selects private key $X_b = 15$
 Evaluate the shared secret key between A and B. Show calculations. 6
- B) List and explain the major features of the AES algorithm. Apply the RSA algorithm to encrypt the message $M = 5$, where $p = 3$, $q = 11$. Compute n , $\phi(n)$, choose $e = 3$, and find the ciphertext C. 6
- Q4. Solve Any Two of the following.
- A) Define the following:
 (a) Hash Function (b) HMAC (c) CMAC (d) Digital Signature
 (e) Message Authentication Code (MAC) (f) Public Key Infrastructure (PKI) 6
- B) Explain the process of message authentication using hash functions. How is it different from encryption? 6
- C) Apply the HMAC algorithm using the following values and show all steps. 6
 Key = 0b10101010,
 Message = '10110011',
 Hash function = simple XOR of key and message.
- Q5. Solve Any Two of the following.
- A) Explain the role and functioning of the SSL/TLS handshake protocol in secure communication. 6
- B) Define the following terms:
 a) SSL b) TLS c) HTTPS d) SSH e) 802.11
 f) Handshake Protocol 6

- C) Explain how the security features of different types of Wi-Fi protections — Open Wi-Fi, WEP, WPA, and WPA2 — are different from each other. Make a table that clearly shows the weaknesses (vulnerabilities) of each one and how the newer versions improved on them. 6

Q6. Solve Any Two of the following.

- A) Define the following terms: 6
a) Pretty Good Privacy (PGP) b) S/MIME c) Authentication Header (AH)
List the components of the IPSec architecture and explain their basic roles in securing IP packets.
- B) Define the following terms: 6
a) Encapsulating Security Payload (ESP) b) Security Association
c) Internet Key Exchange (IKE)
Explain how PGP ensures confidentiality, integrity, and authentication in email communication.
- C) List and briefly explain any three real-world applications of Secure Multiparty Computation (SMC). Explain the working of the Encapsulating Security Payload (ESP) header. What parts of the IP packet does it encrypt and authenticate? 6

*** End ***